PRIYANSHI

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EDUCATION

Indian Institute of Technology (IIT) Kharagpur, India **2018 – 2023**
Dual Degree (B.Tech. Hons. + M.Tech.), Industrial Engineering & Management *CGPA: 7.23/10*
Department of Industrial & Systems Engineering
Master's Thesis: **Ergonomic Evaluation of Healthcare Work Systems in India from the Perspective of Nurses** | *Advisor: Prof. Pradip Kumar Ray*

RESEARCH INTERESTS

Information Systems; Economics of AI; Human–AI Interaction; Future of Work; Digital Innovation; Technology & Society; Computational Social Science.
Methods: causal inference, large-scale digital-trace and text data, NLP/LLM evaluation, panel methods, and machine learning.

WORK-IN-PROGRESS

Geographic Bias in LLMs: Evidence from Reddit Data Accepted - CODE 2026–27
Deepa Mani, Nishant Chadha, and Priyanshi
Extended abstract accepted at the **Conference on the Digital Economy (CODE)**; full manuscript in preparation.

AI Coding Assistants and Technical Debt: A Difference-in-Differences Study Research in progress
with Prof. Vasundhara and Prof. Deepa Mani
Quasi-experimental study of GitHub Copilot and ChatGPT using 471 repositories and approximately 665K commits.

Future of Work in India Research in progress
Large-scale job-posting research on AI/LLMs, occupations, skills, and labour demand in the Indian context.

AI Companies in India: Tracxn / PitchBook Analysis Framing in progress
Company-level empirical analysis of AI firms, startup outcomes, and the Indian AI ecosystem across two independent databases.

RESEARCH EXPERIENCE

Research Associate, Indian School of Business (ISB), Hyderabad **Dec 2024 – Present**
Information Systems | Primary advisor: Prof. Deepa Mani

- Execute end-to-end empirical research workflows spanning research design, literature synthesis, large-scale data construction, statistical analysis, robustness checks, visualization, and academic writing.
- Built and audited LLM-generation pipelines on HPC infrastructure; developed paired India–U.S. Reddit datasets and evaluated geographic differences using sentence embeddings, fixed effects, and clustered inference.
- Designed a difference-in-differences study of AI coding assistants; constructed a panel of **471 GitHub repositories** / **approximately 665,000 commits (2019–2023)** and operationalized technical debt using self-admitted technical debt (SATD).
- Conducted firm-level AI-company analysis using **Tracxn and PitchBook**; work with Revelio and in house-collected Indian job postings to study occupations, skills, and AI exposure in the labour market.
- Replicated prior regressions on **gender and U.S. patent impact** and reconstructed source mappings; separately built Indian-patent research infrastructure covering **654,186 published** and **101,648 granted** patents, with detailed records for **359,391+** publications.
- Developed an initial cloze/question-answer framework for evaluating GPT-based responses in U.S. and Indian startup settings, with attention to factuality, safety, and high-risk-domain evaluation.

COURSEWORK

PhD-level FPM, ISB: Economics A; Economics B; **Econometrics A** (Prof. Bhavana Rai); **Econometrics B** (Prof. Manisha Chakrabarty).
IS / Computational: Information Systems Lab; Software Engineering and Management; Big Data Processing.
Quantitative Methods: Probability & Statistics; Generalized Linear Models & Applications; Game Theory & Applications; Operations Research I–II; Optimisation & Heuristic Methods.

ACADEMIC PROJECTS

Did the US–China Trade War Boost Corporate R&D? Evidence from a Difference-in-Differences Design Econometrics B, ISB | May 2026

PhD-level FPM course | Instructor: Prof. Manisha Chakrabarty | Group project

- Built a Compustat panel for 2015–2022 with **14,641 firm-year observations** / **1,959 firms** and industry-level tariff exposure; estimated two-way fixed-effects DiD models, event-study and placebo specifications, and robustness checks with clustered standard errors.
- Found that R&D-active high-tech firms increased R&D by approximately **9.4 percentage points** more than low-tech peers after the 2018 tariff shock; examined the role of IT/software in the estimated effect.

Replication: Does Women's Labor Force Participation Reduce Domestic Violence? Econometrics A, ISB | 2026

PhD-level FPM course | Instructor: Prof. Bhavana Rai

- Replicated descriptive tables and econometric specifications using Jordan DHS microdata; reconstructed emotional, physical, sexual, and aggregate domestic-violence outcomes and labour-force participation measures.
- Implemented multivariate specifications and endogeneity corrections, including IV/2SLS and 2SRI using a leave-one-out cluster-level female labour-force-participation instrument.

Ergonomic Evaluation of Healthcare Work Systems in India M.Tech Thesis

Advisor: Prof. Pradip Kumar Ray, IIT Kharagpur

- Conducted field-based ergonomic assessment of nurses using questionnaire, anthropometric, and postural data; applied Nordic musculoskeletal measures, Body Part Discomfort measures, and OWAS to identify high-risk activities and intervention opportunities.

Bank Lending Decisions Optimization Mar 2021 – Apr 2021

Advisor: Dr. Akhilesh Kumar, IIT Kharagpur

- Formulated dynamic bank lending as a multi-objective optimization problem balancing profitability and default risk; implemented genetic algorithms, simulated annealing, and a hybrid heuristic and compared solution quality.

Construction Project Scheduling Optimization Aug 2021 – Nov 2021

Advisor: Prof. V. A. Achutha Naikan, IIT Kharagpur

- Applied CPM/PERT to identify critical activities and bottlenecks, analyse time–cost trade-offs, and derive feasible schedules across alternative project-completion dates.

Thief-Tracking Simulation Model Mar 2021 – Apr 2021

- Built an Arena discrete-event simulation of police pursuit and interception under varying traffic conditions; evaluated resource deployment, pursuit success probability, and service levels.

Virtual-Reality Simulation of a Car Interior Jan 2020 – Apr 2020

- Applied anthropometry and work-system-design principles to passenger-car interiors; developed a 3D front-seat/dashboard model in Google SketchUp for later simulation in Unity3D.

SELF-DIRECTED RESEARCH PROJECT

LLMs for Mathematical Optimization: OptiChat Replication [GitHub] 2026 – Present

- Replicated components of an LLM-based optimization assistant using **Pyomo, Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and HiGHS**; implemented LP/MILP test cases and tested feasibility repair, sensitivity, what-if, and counterfactual queries.
- Developed a small evaluation suite based on OptiChat's agent roles to compare open-weight models on structured optimization tasks and diagnose failures in output formatting and counterfactual reasoning.

INDUSTRY EXPERIENCE

Kuehne+Nagel, India **Jul 2023 – Sep 2024**

Graduate Trainee; Assistant Manager, Road Logistics Pricing & Solutions

- Worked across pricing, solution design, and implementation in 3PL/4PL road logistics, supporting PAN-India enterprise transportation operations; developed pricing and forecasting analyses using **XGBoost** and recurrent neural networks.

Akamai Technologies **May 2022 – Jul 2022**

Software Development Engineer Intern

- Worked with distributed infrastructure spanning **3,000+ servers** and more than **1 TB/day** of logs; built EFK-based analytics workflows and applied TF-IDF + K-means clustering.

CONFERENCE

- Dec 2024: Attended ISB–NBER Conference on Information and Competition in the Digital Economy.
- Jan 2025: Attended and volunteered at CODE: Conference on the Digital Economy, ISB.
- Jul 2025: Attended POMS: Conference on the Operations and Supply Chain Management in Emerging Markets, ISB.
- Oct 2025: Attended ISB–Deakin Symposium: Conference on Marketing Technology and AI
- Dec 2025: Attended ISB–NBER Conference on Household Finance across the Lifecycle.